

SOLARIS SPEED RUN

Transition to Solaris in An Hour

Andreas Weidman

Swiss International, Stockholm, Sweden

andreas@swiss.se

ABSTRACT

This hands-on class breaks down the steps that a small 30 person studio took to transition to SideFX Solaris. Solaris is a suite of look development, layout and lighting tools that empower artists to create USD based scene graphs that go from asset creation to final render.

Solaris integrates with USD's HYDRA Imaging Framework for access to a wide range of renderers such as the new SideFX Karma,* Pixar RenderMan, Autodesk Arnold, Maxon Redshift, AMD ProRender and more.

In this class we will simplify Solaris, the pipeline, and USD concepts. We will discuss practical tips & tricks to guide how our small studio transitioned to Solaris as our daily driver. We will help you look at how you think about Solaris, what to build; and the practical steps getting a Solaris project done from start to finish.

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1 OVERALL CLASS AGENDA

1.1 Solaris intro & benefits overview - Recent updates to Solaris

A short intro to what Solaris is and the use for it when it comes to lookdev, lightning and rendering. While not diving into each point, broad use cases are quickly mentioned as well as the recent updates listed below.

1.1.1 Solaris for Layout and Environments.

- Unique brush-based framework for large-scale set dressing.
- HDA-driven brush architecture with unlimited customizability.
- Full access to verified/compliable SOP networks.
- PDG-assisted creation of layout asset gallery.
- USD asset browser.

1.1.2 Solaris for Lighting and Rendering.

- Comprehensive Houdini scene importer.
- Better motion blur generation.

- Option to pause viewer updates.
- Faster GL rendering of transformable instances.
- Viewport support for camera exposure attribute.
- Updated Karma LOP for faster render setup.
- USD 21.08.

1.2 Simplifying Solaris & USD concepts

In this class we will break down and simplify the USD and Solaris workflow. The USD image workflow is typically described by larger studios that have custom tools sets and workflows that may be less relevant to the smaller studios or teams. We are going to talk a little about USD and workflows in general, and we will also cover using Solaris without USD. We will show our workflow with real world examples as well as showing how we could do the effects without the use of a lot of USD pipeline.

- Building a manifest & naming conventions - Going through the USD-concept of manifests to control how things are named across shots and sequences for a unified workflow between artist departments.
- Setting up work environment & contexts - To utilize the multi-shot workflows, we are able to set up different contexts and variables that can change dynamically per shot, like render paths for instance.
- Working with assets & SOPs - How to create things within Solaris or bringing in external assets. How to apply shading and how to structure it.
- Dealing with sublayers - Explaining the concept of using usd files as layers within the scene and the files themselves, in order to minimize export/import times and filesize.
- Multi-shot workflow strategies - How to utilize and reuse lights, settings, assets and other things when working with multiple shots to save time and efforts.
- Rendering strategies - How to leverage rendersettings and optimize for both IPR rendering as well as farm rendering.
- End thoughts and conclusion.

2 CONCLUSION

We had two main take aways from building our tool set. The first being our tool set is it must be 100% haxable, Since we work in commercials, we do not deal with a lot of shots and assets. Almost every project is something you've never done before and you don't have a lot of time to complete it. You also must be able to move very freely.

The second thing was to eliminate time wasters like opening a scene, publishing, and updating a scene. We worked with our artists to figure out what they spend time on through out the day and we minimized each step so that our artist or freelancers could

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jump in easily and get up and running in just a few hours. We also wanted to make sure it played well with vanilla nodes of Solaris.

The key take away from this class is to not worry too much about USD and the workflows, but to start seeing Solaris as a playground. There are no rules that you must stick by and you should experiment with what works best for you.

3 SPEAKER BACKGROUND & STUDIO INTRO

Andreas Weidman is a Visual Effects Supervisor based in Sweden, with over 20-years experience ranging from Art Direction, Concept Art and Motion Design, to the full spectrum of Visual Effects. Having spent half a decade at Important Looking Pirates working on TV-shows like Mandalorian, Westworld, and Fear the Walking

Dead; he is now directing his focus on VFX-commercials at the award winning studio Swiss International.

4 METHOD OF PRESENTATION

Primarily a show-and-tell projected, with the option for attendees to follow along during the presentation or afterwords. While fast paced, it is the concepts themselves that will be broken down and simplified for attendees to utilise and internalize.

5 INTENDED AUDIENCE

While the class will dive into some complex details, this talk will help anyone interested in USD understand the concept a bit more. Most beneficial will be for experienced 3D users and pipeline developers, as well as Houdini users.